SHRI MAHAPRABHU PUBLIC SCHOOL

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PRACTICAL PROGRAMS LIST — SESSION 2026–27

CLASS: XII | SUBJECT: COMPUTER SCIENCE

SECTION A: PYTHON PROGRAMMING & FILE HANDLING

1. Write a Python program to calculate the income tax of an employee based on the basic salary and total savings inputted by the user according to the following slabs:

- Income < ₹ 2,50,000: No tax
- ₹ 2,50,000 to ₹ 5,00,000: 5%
- ₹ 5,00,000 to ₹ 10,00,000: 20%
- Income > ₹ 10,00,000: 30%

(Investments up to ₹ 1,50,000 under Sec 80C allow tax savings up to ₹ 45,000).

2. Write a Python program to input a string and count the number of uppercase and lowercase letters.

3. Write a Python program to find the maximum, minimum, and mean values from a user-inputted list.

4. Write a Python program to generate the Fibonacci series up to N terms and store it in a tuple.

5. Write a Python program to store student details (Admission Number, Roll Number, Name, Marks) in a dictionary and display a student's information based on their Admission Number.

6. Write a Python program to modify a given string such that its first and last characters are swapped.

7. Write a Python program to input details for N customers (Items bought, Cost, Phone Number), store them in a nested dictionary, and display all details in a clean tabular format.

8. Write a Python program that accepts a list of integers from the user and creates a new list containing only the numbers having three or more digits.

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Example:
Existing List: [10, 100, 99, 200, 1000]
New List: [100, 200, 1000]
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9. Write a Python program that increases the numeric value by 10 for dictionary items whose keys are alphabetic vowels.

10. Write a Python program that accepts a list Arr and an integer n, shifting all elements of the list to the left by n positions.

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Sample Input: Arr = [10, 20, 30, 40, 12, 11], n = 2
Output: Arr = [30, 40, 12, 11, 10, 20]
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11. Write a user-defined function that accepts a string as input and counts/displays the total frequency of each character.

12. Write a user-defined function to accept elements into a tuple and display the count of even and odd numbers present in it.

13. Write a user-defined function prime_numbers(N) in Python to find and display all prime numbers between 2 and N.

14. Write a user-defined function power(x, y=2) to compute x^y. If y is not provided, compute x^2.

15. Write a function stats(filename) that reads a text file and reports its longest line.

16. Write a program that reads an alphanumeric text file (an.txt) and prints only the numeric digits.

17. Write a program that copies lines from source.txt to target.txt, skipping any lines that begin with '@'.

18. Write a program to read a text file line-by-line and display each word separated by '#'.

19. Write a program to display all records/lines in a file along with their corresponding line/record numbers.

20. A binary file STOCK.DAT has the structure [ITEMID, ITEMNAME, QUANTITY, PRICE]. Write a user-defined function MakeFile() to input a record and append it to STOCK.DAT.

21. A binary file EMP.dat has the structure [EmpNo, EName, Post, Salary]. Write a user-defined function NewEmp() to append new employee records.

22. A binary file TEST.dat contains [TestId, Subject, MaxMarks, ScoredMarks]. Write a function DisplayAvgMarks(Sub) that calculates and displays the average marks scored in a specified subject.

23. Write a Python program to search and display student records (Rollno, Name, Marks) from a binary file Student.dat based on a user-entered Roll Number.

24. Write a program to write Employee records (EmpNo, EName, Salary) into a CSV file.

25. Write user-defined functions to perform read and write operations on student.csv containing fields [RollNo, Name, Stream, Marks].

26. Write a menu-driven program with functions to:

a) Write multiple records to student.csv at once.

b) Display all contents of student.csv.

27. Write a program to read data.txt and count the total number of blank spaces in it.

28. Write a program to count occurrences of the word 'the' (case-insensitive) in article.txt.

29. Write a function COUNTLINES_ET() to read REPORT.TXT and count how many lines start with 'E' and how many start with 'T', displaying both counts separately.

30. Write a program to read inventory.csv (columns: Product, Quantity, Price) using csv.DictReader(). Calculate the total stock value (Quantity * Price) for each product and print a summary report.

SECTION B: DATA STRUCTURES (STACKS)

31. Write a function Push(SItem) where SItem is a dictionary {Sname: Price}. Push item names into a stack if Price > 75 and display the count of elements pushed.

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Example Input: Ditem = {"Pen": 106, "Pencil": 59, "Notebook": 80, "Eraser": 25}
Stack Content: ['Notebook', 'Pen']
Output: The count of elements in the stack is 2
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32. Given a dictionary R = {"OM": 76, "JAI": 45, "BOB": 89, "ALI": 65, "ANU": 90, "TOM": 82}:

- Push the keys (student names) into a stack where corresponding marks > 75.
- Pop and display all content from the stack.

```
Expected Output: TOM ANU BOB OM
```

33. Given a list N = [12, 13, 34, 56, 21, 79, 98, 22, 35, 38]:

- Traverse the list and push all even numbers into a stack.
- Pop and display the content of the stack.

```
Expected Output: 38 22 98 56 34 12
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SECTION C: STRUCTURED QUERY LANGUAGE (SQL)

34. Consider the tables SCHOOL and ADMIN to write the following SQL queries:

TABLE: SCHOOL

CODE	TEACHERNAME	SUBJECT	DOJ	PERIODS	EXPERIENCE
1001	RAVI SHANKAR	ENGLISH	12/03/2000	24	10
1009	PRIYA RAI	PHYSICS	03/09/1998	26	12
1203	LISA ANAND	ENGLISH	09/04/2000	27	5
1045	YASHRAJ	MATHS	24/08/2000	24	15
1123	GANAN	PHYSICS	16/07/1999	28	3
1167	HARISH B	CHEMISTRY	19/10/1999	27	5
1215	UMESH	PHYSICS	11/05/1998	22	16

TABLE: ADMIN

CODE	GENDER	DESIGNATION
1001	MALE	VICE PRINCIPAL
1009	FEMALE	COORDINATOR
1203	FEMALE	COORDINATOR
1045	MALE	HOD
1123	MALE	SENIOR TEACHER
1167	MALE	SENIOR TEACHER
1215	MALE	HOD

i. Display total PERIODS for each SUBJECT from SCHOOL table.
ii. Display TEACHERNAME, GENDER from tables SCHOOL and ADMIN whose DESIGNATION is 'COORDINATOR'.
iii. Display TEACHERNAME and DOJ in descending order of CODE.
iv. Display TEACHERNAME whose DOJ is in the year 2000.

35. Write SQL statements for the following queries based on relations CUSTOMER (ACNO, NAME, GENDER, BALANCE) and TRANSACTION (ACNO, TDATE, AMOUNT, TYPE):

Table: CUSTOMER

ACNO	NAME	GENDER	BALANCE
C1	RISHABH	M	15000
C2	AAKASH	M	12500
C3	INDIRA	F	9750
C4	TUSHAR	M	14600
C5	ANKITA	F	22000

Table: TRANSACTION

ACNO	TDATE	AMOUNT	TYPE
C1	2020-07-21	1000	DEBIT
C5	2019-12-31	1500	CREDIT
C3	2020-01-01	2000	CREDIT

i. To display all information about CUSTOMERS whose NAME starts with 'A'.
ii. To display NAME and BALANCE of Female CUSTOMERS (GENDER = 'F') whose TRANSACTION Date (TDATE) is in the year 2019.
iii. To display the total number of CUSTOMERS for each GENDER.
iv. To display CUSTOMER NAME and BALANCE in ascending order of GENDER.
v. To display CUSTOMER NAME and their respective INTEREST for all CUSTOMERS where INTEREST is calculated as 8% of BALANCE.

36. Write SQL queries for (i) to (iv) and find outputs for (v) to (viii) based on tables DVD (DCODE, DTITLE, DTYPE) and MEMBER (MID, NAME, DCODE, ISSUEDATE):

DVD

DCODE	DTITLE	DTYPE
F101	Henry Martin	Folk
C102	Dhrupad	Classical
C101	The Planets	Classical
F102	Universal Soldier	Folk
R102	A day in life	Rock

MEMBER

MID	NAME	DCODE	ISSUEDATE
101	AGAM SINGH	R102	2017-11-30
103	ARTH JOSEPH	F102	2016-12-13
102	NISHA HANS	C101	2017-07-24

i. Display all details from table MEMBER in descending order of ISSUEDATE.
ii. Display DCODE and DTITLE of all Folk Type DVDs from table DVD.
iii. Display DTYPE and number of DVDs in each DTYPE from table DVD.
iv. Display NAME and ISSUEDATE of members who have DVDs issued in the year 2017.

v. SELECT MIN(ISSUEDATE) FROM MEMBER;
vi. SELECT DISTINCT DTYPE FROM DVD;
vii. SELECT D.DCODE, NAME, DTITLE FROM DVD D, MEMBER M WHERE D.DCODE = M.DCODE;
viii. SELECT DTITLE FROM DVD WHERE DTYPE NOT IN ('Folk', 'Classical');

37. Write SQL queries for (i) to (iv) and find outputs for (v) to (viii) based on tables VEHICLE (VCODE, VEHICLETYPE, PERKM) and TRAVEL (CNO, CNAME, TRAVELDATE, KM, VCODE, NOP):

Table : VEHICLE

VCODE	VEHICLETYPE	PERKM
V01	VOLVO BUS	150
V02	AC DELUXE BUS	125
V03	ORDINARY BUS	80
V05	SUV	30
V04	CAR	18

NOTE PERKM is Freight Charges per kilometre

Table : TRAVEL

CNO	CNAME	TRAVELDATE	KM	VCODE	NOP
101	K. Niwal	2015-12-13	200	V01	32
103	Fredrick Sym	2016-03-21	120	V03	45
105	Hitesh Jain	2016-04-23	450	V02	42
102	Ravi Anish	2016-01-13	80	V02	40
107	John Malina	2015-02-10	65	V04	2
104	Sahanubhuti	2016-01-28	90	V05	4
106	Ramesh Jaya	2016-04-06	100	V01	25

NOTE

- KM is Kilometres travelled
- NOP is number of passengers travelled in vehicle.

i. Display CNO, CNAME, TRAVELDATE from table TRAVEL in descending order of CNO.
ii. Display CNAME of customers traveling by vehicle with code V01 or V02.
iii. Display CNO and CNAME of customers who travelled between '2015-05-01' and '2015-12-31'.
iv. Display all details from table TRAVEL for customers having travel distance > 120 KM in ascending order of NOP.
v. SELECT COUNT(*), VCODE FROM TRAVEL GROUP BY VCODE HAVING COUNT(*) > 1;
vi. SELECT DISTINCT VCODE FROM TRAVEL;
vii. SELECT A.VCODE, CNAME, VEHICLETYPE FROM TRAVEL A, VEHICLE B WHERE A.VCODE = B.VCODE AND KM < 90;
viii. SELECT CNAME, KM * PERKM FROM TRAVEL A, VEHICLE B WHERE A.VCODE = B.VCODE AND A.VCODE = 'V05';

38. Write SQL queries for (i) to (iv) and find outputs for (v) to (viii) based on tables DEPT (DCODE, DEPARTMENT, CITY) and WORKER (WNO, NAME, DOJ, DOB, GENDER, DCODE):

Table : DEPT

DCODE	DEPARTMENT	CITY
D01	MEDIA	DELHI
D02	MARKETING	DELHI
D03	INFRASTRUCTURE	MUMBAI
D05	FINANCE	KOLKATA
D04	HUMAN RESOURCE	MUMBAI

Table : WORKER

WNO	NAME	DOJ	DOB	GENDER	DCODE
1001	George K	2013-09-02	1991-09-01	MALE	D01
1002	Ryma Sen	2012-12-11	1990-12-15	FEMALE	D03
1003	Mohitesh	2013-02-03	1987-09-04	MALE	D05
1007	Anil Jha	2014-01-17	1984-10-19	MALE	D04
1004	Manila Sahai	2012-12-09	1986-11-14	FEMALE	D01
1005	R SAHAY	2013-11-18	1987-03-31	MALE	D02
1006	Jaya Priya	2014-06-09	1985-06-23	FEMALE	D05

i. Display WNO, NAME, GENDER from table WORKER in descending order of WNO.

ii. Display NAME of all FEMALE workers from table WORKER.

iii. Display WNO and NAME of workers born between '1987-01-01' and '1991-12-01'.

iv. Count and display MALE workers who joined after '1986-01-01'.

v. SELECT COUNT(*), DCODE FROM WORKER GROUP BY DCODE HAVING COUNT(*) > 1;

vi. SELECT DISTINCT DEPARTMENT FROM DEPT;

vii. SELECT NAME, DEPARTMENT, CITY FROM WORKER W, DEPT D WHERE W.DCODE = D.DCODE AND WNO < 1003;

viii. SELECT MAX(DOJ), MIN(DOB) FROM WORKER;

39. Answer questions (a) and (b) based on tables STORE (SNo, SName, Area) and ITEM (INo, IName, Price, SNo):

TABLE : STORE

SNo	SName	Area
S01	ABC Computronics	GK II
S02	All Infotech Media	CP
S03	Tech Shoppe	Nehru Place
S04	Geeks Tecno Soft	Nehru Place
S05	Hitech Tech Store	CP

TABLE : ITEM

INo	IName	Price	SNo
T01	Mother Board	12000	S01
T02	Hard Disk	5000	S01
T03	Keyboard	500	S02
T04	Mouse	300	S01
T05	Mother Board	13000	S02
T06	Keyboard	400	S03
T07	LCD	6000	S04
T08	LCD	5500	S05
T09	Mouse	350	S05
T10	Hard Disk	4500	S03

a) Write SQL queries:

i. Display IName and Price of all items in ascending order of Price.

ii. Display SNo and SName of all stores located in 'CP'.

iii. Display Minimum and Maximum Price of each IName from table ITEM.

iv. Display IName, Price of all items and their respective SName where available.

b) Write outputs of following SQL commands:

i. SELECT DISTINCT IName FROM ITEM WHERE Price >= 5000;

ii. SELECT Area, COUNT(*) FROM STORE GROUP BY Area;

iii. SELECT COUNT(DISTINCT Area) FROM STORE;

iv. SELECT IName, Price * 0.05 AS DISCOUNT FROM ITEM WHERE SNo IN ('S02', 'S03');

SECTION D: PYTHON-MYSQL DATABASE CONNECTIVITY

40. Write a Python connectivity program to fetch and display only the first three records from a database table using fetchmany(3).

41. Write a Python connectivity program to display records of all female managers from table WORKER and find out the total number of female managers. If no records exist, print 'Record Not Found'.

Table : DEPT

DCODE	DEPARTMENT	CITY
D01	MEDIA	DELHI
D02	MARKETING	DELHI
D03	INFRASTRUCTURE	MUMBAI
D05	FINANCE	KOLKATA
D04	HUMAN RESOURCE	MUMBAI

Table : WORKER

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1003	Mohitesh	2013-02-03	1987-09-04	MALE	D05
1007	Anil Jha	2014-01-17	1984-10-19	MALE	D04
1004	Manila Sahai	2012-12-09	1986-11-14	FEMALE	D01
1005	R SAHAY	2013-11-18	1987-03-31	MALE	D02
1006	Jaya Priya	2014-06-09	1985-06-23	FEMALE	D05

42. Write a Python connectivity program to display records of all managers from table WORKER using a parameterized query (%s placeholders) and count total managers. If details do not exist, print 'Record Not Found'.

43. Write the complete steps and Python program to perform an INSERT query in database connectivity. Table Student has columns (rollno, name, age) and values to insert are (10, 'Ashok', 26). Ensure transaction is committed.

44. Write a Python function that accepts name as a parameter and searches table Student (rollno, name, age) to determine whether the record exists or not.

45. A table named STATIONERY in database ITEMDB has structure: [itemNo int(11), itemName varchar(15), price float, qty int(11)].

Field	Type
itemNo	int(11)
itemName	varchar(15)
price	float
qty	int(11)

Write a Python function AddAndDisplay() to input item details, store them in table STATIONERY, and then retrieve and display all records where Price > 120.
Assume Python-Database connectivity parameters: Host: localhost, User: root, Password: Pencil.